

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 3: Calculation of CG & % MAC (Mean Aerodynamic Chord)

Syllabus Source: CAE Oxford (pp. 43-66) | Pass Target: >= 90% AviationExam

1. THE MEAN AERODYNAMIC CHORD (MAC) CONCEPT

Because modern aircraft have swept, tapered wings, the aerodynamic chord varies along the wingspan. The Mean Aerodynamic Chord (MAC / $c_{\bar{}}$) is the chord of an equivalent rectangular wing that has the identical pitching moment and aerodynamic characteristics as the actual wing!

- LEMAC: Leading Edge of the Mean Aerodynamic Chord. The distance from the reference Datum to the front edge of the MAC.
- TEMAC: Trailing Edge of the Mean Aerodynamic Chord. LEMAC + Length of MAC = TEMAC.
- Why Express CG as % MAC?: Expressing CG as a percentage of MAC allows pilots to understand the aerodynamic stability and trim of an aircraft immediately, regardless of whether the aircraft is a small trainer or a heavy Boeing 777!

2. THE MASTER CONVERSION FORMULAS: DISTANCE <-> % MAC

- Formula 1: Converting CG Distance into % MAC: $\% \text{ MAC} = ((\text{CG Distance from Datum} - \text{LEMAC}) / \text{Length of MAC}) \times 100$
- Formula 2: Converting % MAC into CG Distance from Datum: $\text{CG Distance from Datum} = \text{LEMAC} + (\% \text{ MAC} / 100) \times \text{Length of MAC}$

3. STABILITY & CONTROLLABILITY TRADE-OFF AT THE LIMITS

CG Position Regime	Aerodynamic Pitch Stability & Stick Forces	Flight Control Authority & Fuel Performance
CG Position Regime	Aerodynamic Pitch Stability & Stick Forces	Flight Control Authority & Fuel Performance
Forward CG Limit	MAXIMUM static longitudinal stability. Very heavy stick forces (high stick force per g).	Landing flare authority severely limited! High downward tailplane force -> high induced trim drag and higher fuel burn.
Aft CG Limit	MINIMUM static stability. Light, sensitive stick forces. Danger of over-controlling.	Lower tailplane downforce -> LOWER TRIM DRAG, lower fuel burn. Must NOT exceed Neutral Point (neutral/unstable!).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): CG as % MAC Direct & Inverse C

Problem 1 (Distance to % MAC): LEMAC = 14.0 m, TEMAC = 18.0 m. Aircraft CG is 15.2 m aft of datum. - Step 1: MAC Length = TEMAC - LEMAC = 18.0 - 14.0 = 4.0 meters. - Step 2: Distance from LEMAC = CG - LEMAC = 15.2 - 14.0 = 1.2 meters. - Step 3: % MAC = $(1.2 / 4.0) \times 100 = 30\% \text{ MAC}$. Problem 2 (Inverse: % MAC to Distance): If CG is 25% MAC with LEMAC = 12.0 m and MAC = 5.0 m: - Formula: $\text{CG} = \text{LEMAC} + (\% \text{ MAC} / 100) \times \text{MAC}$ = $12.0 + (0.25 \times 5.0) = 13.25 \text{ meters aft of datum}$.

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Converting CG Position to % MAC and Vice Versa

SCENARIO 1 (Finding % MAC): Given the following aircraft technical data: - LEMAC is located at: 14.0 meters aft of datum - Length of the MAC: 4.0 meters - Actual aircraft CG calculated from loading sheet: 15.2 meters aft of datum QUESTION: What is the aircraft's Center of Gravity expressed as a percentage of MAC (% MAC)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the distance from LEMAC to the CG: - Distance from LEMAC = CG - LEMAC = 15.2 m - 14.0 m = 1.2 meters. Step 2: Divide this distance by the total length of the MAC: - Fraction of MAC = 1.2 m / 4.0 m = 0.30. Step 3: Multiply by 100 to get a percentage: - % MAC = 0.30 x 100 = 30.0% MAC. FINAL ANSWER: CG = 30.0%

MAC. ----- SCENARIO

2 (Finding Physical CG Distance from % MAC): QUESTION: If the certified forward limit is 18% MAC, at what distance from the datum is the forward limit located? SOLUTION PASO A PASO: Step 1: Convert percentage to decimal: 18% = 0.18. Step 2: Multiply decimal by MAC length: 0.18 x 4.0 m = 0.72 meters aft of LEMAC. Step 3: Add this distance to LEMAC: 14.0 m + 0.72 m = 14.72 meters. FINAL ANSWER: Forward CG Limit is at 14.72 m aft of Datum.

[!] EXAM TRAP (AVIATIONEXAM) - TEMAC and MAC Length Trap

- If an exam question gives you LEMAC and TEMAC instead of MAC length: - Formula: Length of MAC = TEMAC - LEMAC! - E.g. If LEMAC = 500 inches and TEMAC = 700 inches, the MAC length is 200 inches.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: If LEMAC is at station 20 m and TEMAC is at station 25 m, what is the length of the Mean Aerodynamic Chord?

- [A]: 45 m
- [B]: 5 m
- [C]: 22.5 m
- [D]: 10 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. MAC Length = TEMAC - LEMAC = 25 m - 20 m = 5 meters.

Q2: An aircraft has LEMAC at 1,000 inches and MAC length of 150 inches. If the CG is at 26% MAC, what is the CG position?

- [A]: 1,039 inches
- [B]: 1,026 inches
- [C]: 1,150 inches
- [D]: 1,015 inches

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. CG = LEMAC + (% MAC / 100 x MAC) = 1,000 + (0.26 x 150) = 1,000 + 39 = 1,039 inches.

Q3: Why does flying with a Center of Gravity at the AFT limit result in lower fuel consumption than at the FORWARD limit?

- [A]: The engines produce more thrust
- [B]: Less tailplane download is required to balance the aircraft, which reduces total lift required and lowers induced trim drag

- [C]: The aircraft flies at lower TAS
- [D]: Air density increases

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At an aft CG, the nose-down pitching moment is small, requiring a smaller downward balancing force from the horizontal stabilizer. Lower total lift required means lower induced trim drag, saving fuel.